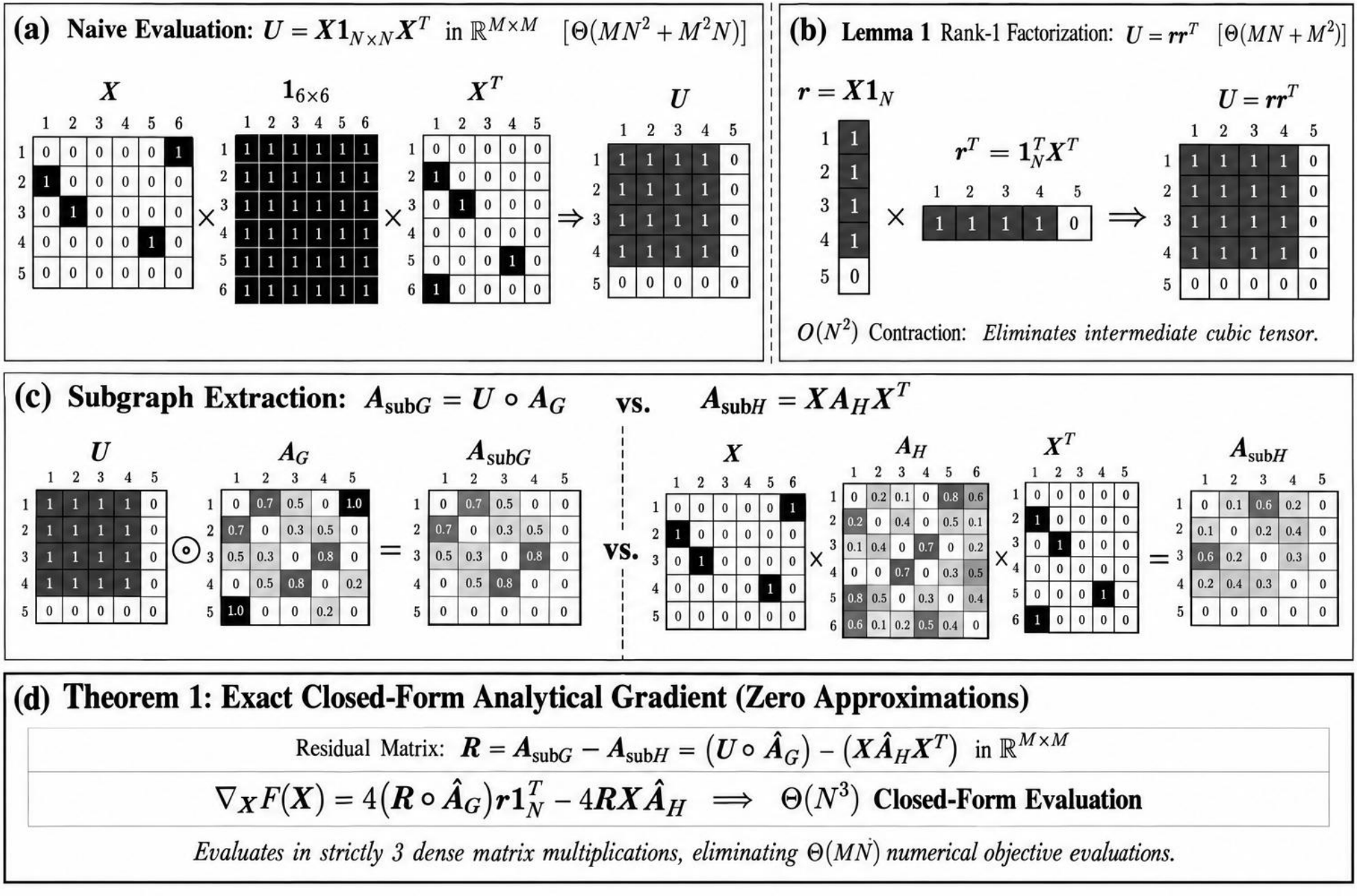

(a) Naive Evaluation: $U = X1_{N\times N}X^T$ in $\mathbb{R}^{M\times M}$ $[\Theta(MN^2 + M^2N)]$
X
$1_{6\times6}$
X^T
U
(b) Lemma 1 Rank-1 Factorization: $U = rr^T$ $[\Theta(MN + M^2)]$
$r = X1_N$
$r^T = 1_N^T X^T$
$U = rr^T$
$O(N^2)$ Contraction: Eliminates intermediate cubic tensor.
(c) Subgraph Extraction: $A_{\text{sub}G} = U \circ A_G$ vs. $A_{\text{sub}H} = XA_HX^T$
U
A_G
$A_{\text{sub}G}$
vs.
X
A_H
X^T
$A_{\text{sub}H}$
(d) Theorem 1: Exact Closed-Form Analytical Gradient (Zero Approximations)
Residual Matrix: $R = A_{\text{sub}G} - A_{\text{sub}H} = \left(U \circ \hat{A}_G\right) - \left(X\hat{A}_HX^T\right)$ in $\mathbb{R}^{M\times M}$
$\nabla_X F(X) = 4\left(R \circ \hat{A}_G\right)r1_N^T - 4RX\hat{A}_H \implies \Theta(N^3)$ Closed-Form Evaluation
Evaluates in strictly 3 dense matrix multiplications, eliminating $\Theta(MN)$ numerical objective evaluations.